RAJALAKSHMI ENGINEERING COLLEGE

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L2

Started on	Monday, 31 August 2026, 9:29 PM
State	Finished
Completed on	Monday, 31 August 2026, 9:38 PM
Time taken	9 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int count(int a[], int l, int r, int x) {
    int c = 0;
    for (int i = l; i <= r; i++) {
        if (a[i] == x) c++;
    }
    return c;
}
int solve(int a[], int l, int r) {
    if (l == r) return a[l];
    int m = (l + r) / 2;
    int lc = solve(a, l, m);
    int rc = solve(a, m + 1, r);
    if (lc == rc) return lc;
    int countL = count(a, l, r, lc);
    int countR = count(a, l, r, rc);
    return (countL > countR) ? lc : rc;
}
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    printf("%d\n", solve(a, 0, n - 1));
    return 0;
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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